Appendix C HARDWARE Results

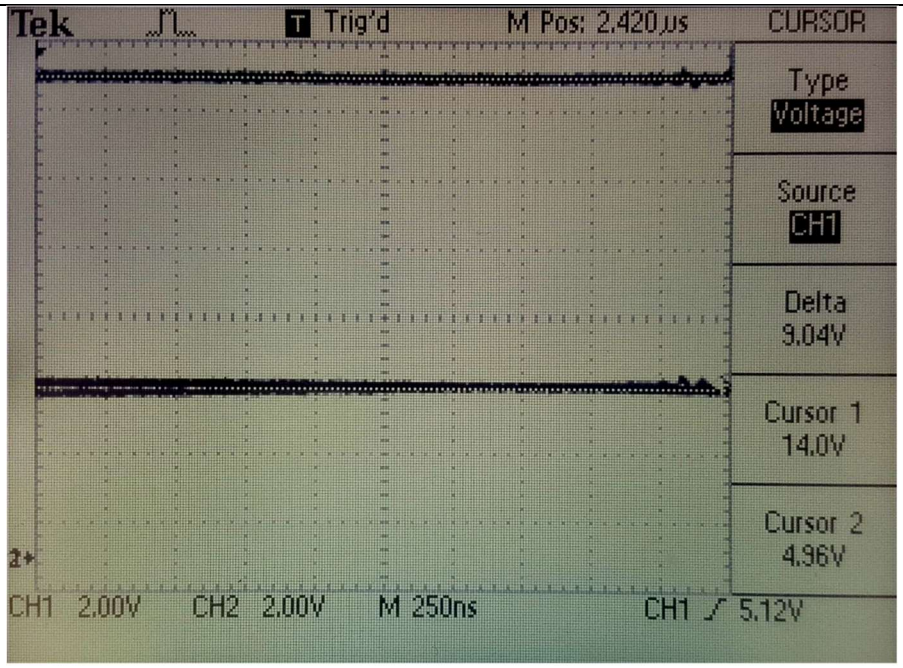

Fig. 1 Output Voltage at 14V input.

Fig. 1 shows the Buck's output voltage with an input voltage of 14V. The output voltage is 4.96V as shown in "Cursor 2" block with an input voltage of 14V as shown in "Cursor 1" block.

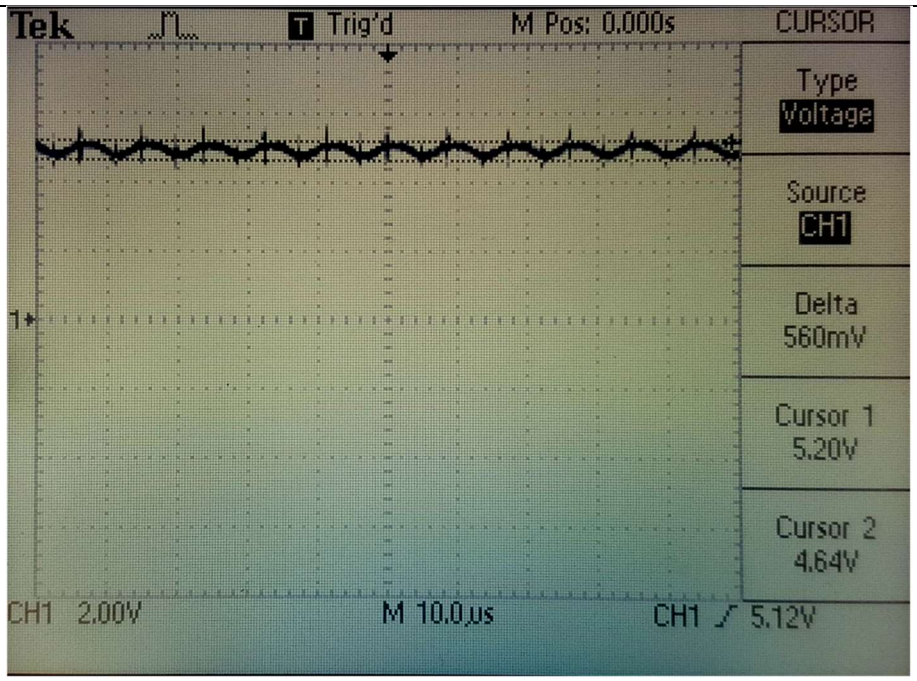

Fig. 2 Output Voltage Ripple at 14V input

Fig. 2 shows the output voltage ripple at an input voltage of 14V. The maximum output voltage is 5.20V as shown in “Cursor 1” block and the minimum output voltage is 4.64V as shown in “Cursor 2” block at an input voltage of 14V.

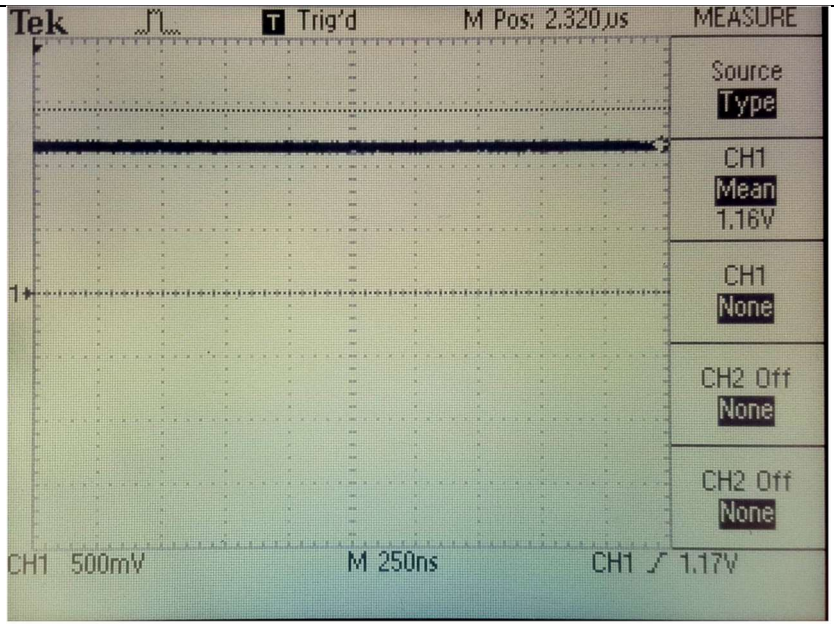

Fig. 3 Controller output voltage at 14V input

Fig. 3 shows the controller output voltage at an input voltage of 14V. The controller output voltage is 1.16V as shown in the "Mean" block at an input voltage of 14V.

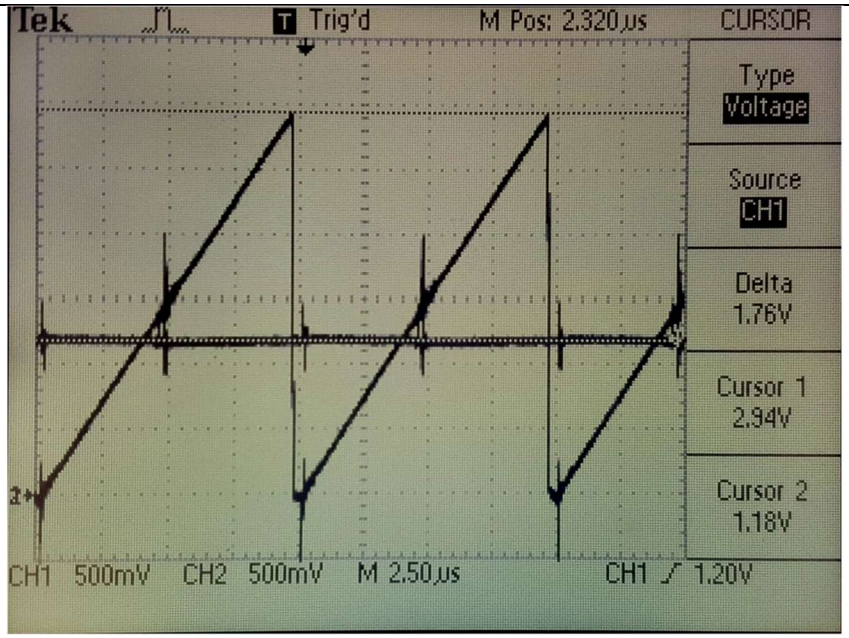

Fig. 4Saw tooth signal and controller output voltage at 14V input

Fig. 4 shows the saw tooth signal and the controller output voltage at an input voltage of 14V. The saw tooth's voltage is 2.94V as shown in "Cursor 1" block and the controller output voltage is 1.18V as shown in "Cursor 2" block, at an input voltage of 14V.

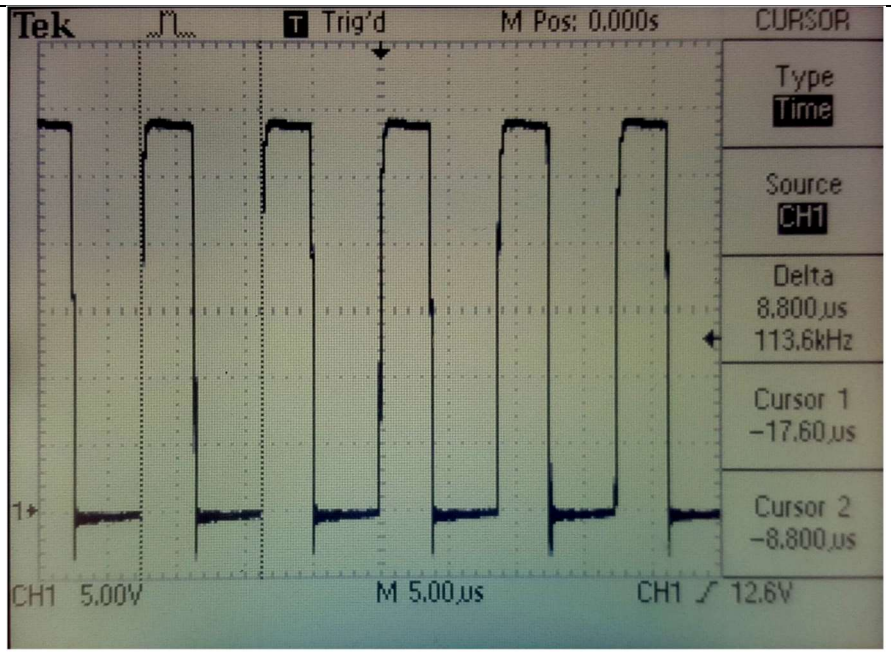

Fig. 5 Period of switching square wave at 14V input

Fig. 5 shows the switching square wave at an input voltage of 14V. The figure shows the period of the switching square wave shown in “Cursor 1” and “Cursor 2” blocks, at an input voltage of 14V.

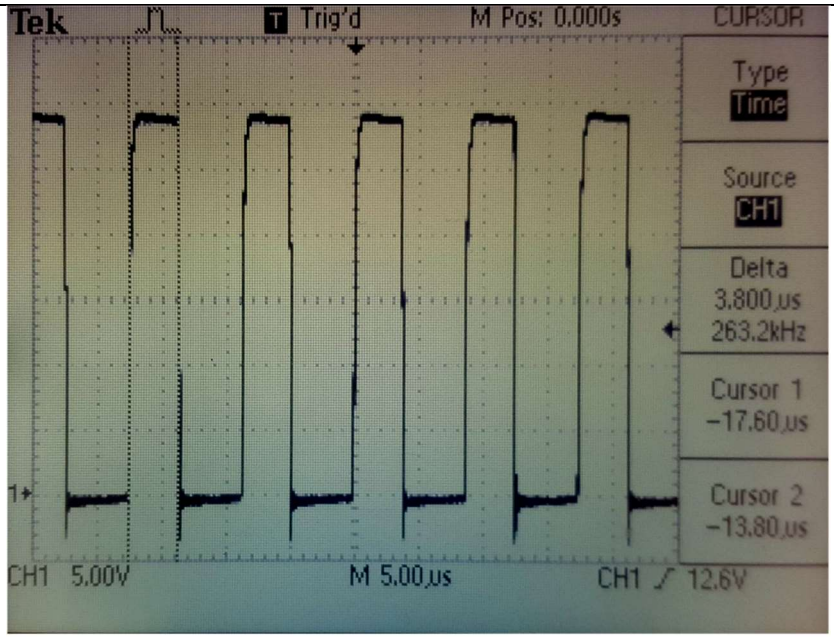

Fig. 6 On time of switching square wave at 14V input

Fig. 6 shows the switching square wave at an input voltage of 14V. The figure shows the on the time of the switching square wave shown in “Cursor 1” and “Cursor 2” blocks, at an input voltage of 14V.

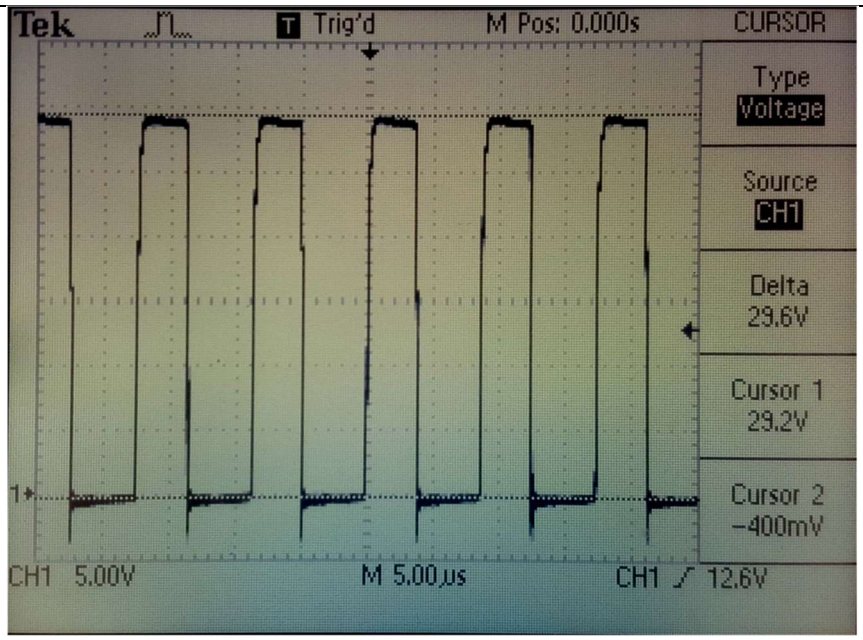

Fig. 7 Switching square wave amplitude at 14V input

Fig. 7 shows the switching square wave at an input voltage of 14V. The figure shows the amplitude of the switching square wave shown in “Cursor 1” and “Cursor 2” blocks, at an input voltage of 14V.

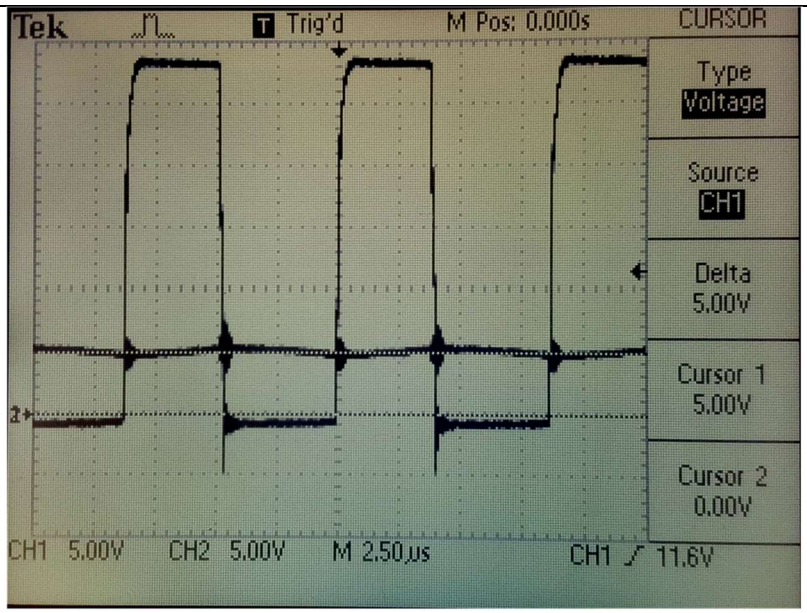

Fig. 8 Control voltage and switching square wave at 14V input

Fig. 8 shows the control voltage signal and switching frequency square wave at an input voltage of 14V. To show the duty cycle with the amplitude of the control voltage which is 5V as shown in “Cursor 2” block, at an input voltage of 14V.

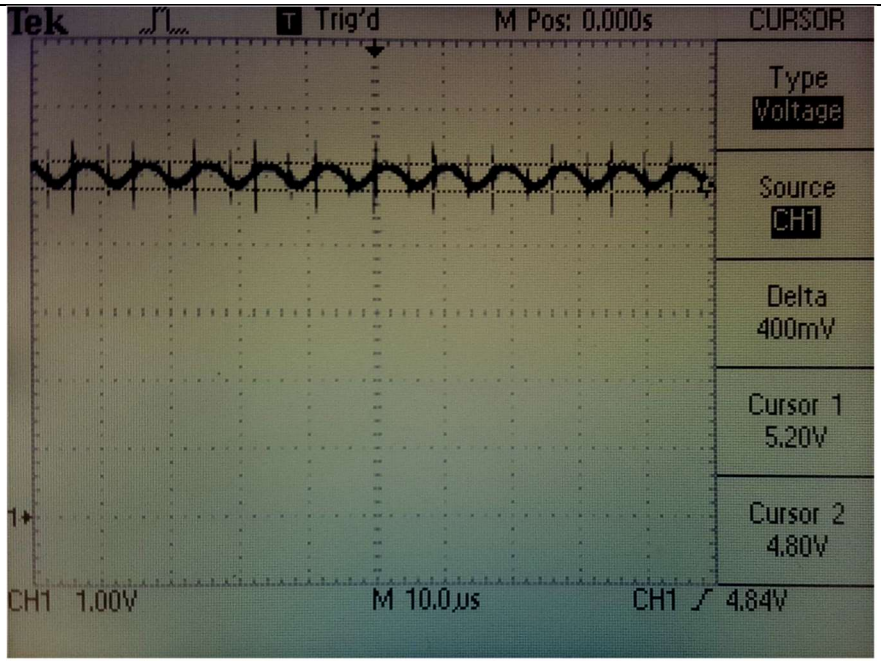

Fig. 9 Output Voltage Ripple at 14.5V input

Fig. 9 shows the output voltage ripple at an input voltage of 14.5V. The maximum output voltage is 5.20V as shown in “Cursor 1” block and the minimum output voltage is 4.8V as shown in “Cursor 2” block at an input voltage of 14.5V.

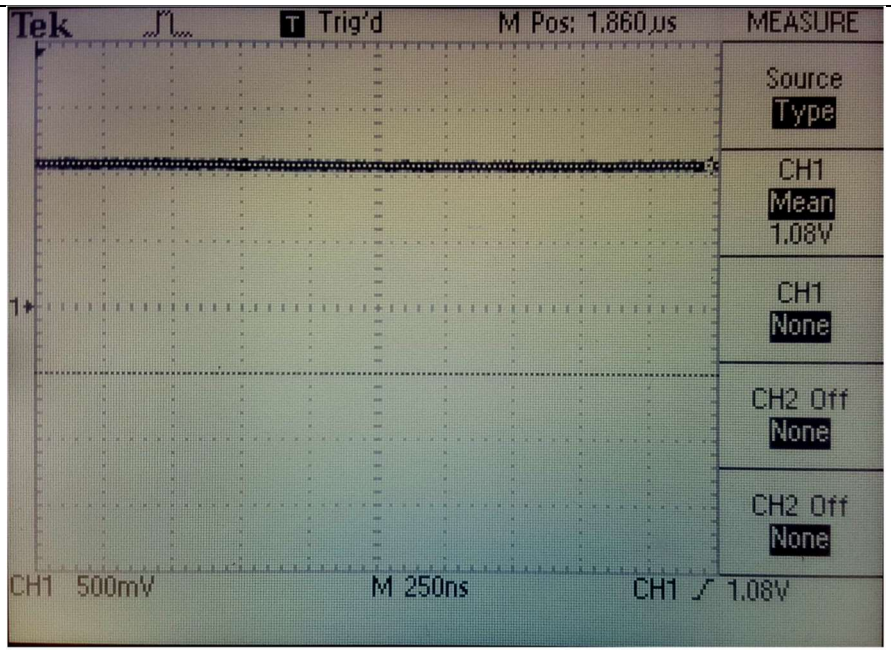

Fig. 10 Controller output voltage at 14.5V input

Fig. 10 shows the controller output voltage at an input voltage of 14.5V. The controller output voltage is 1.08V as shown in the "Mean" block.

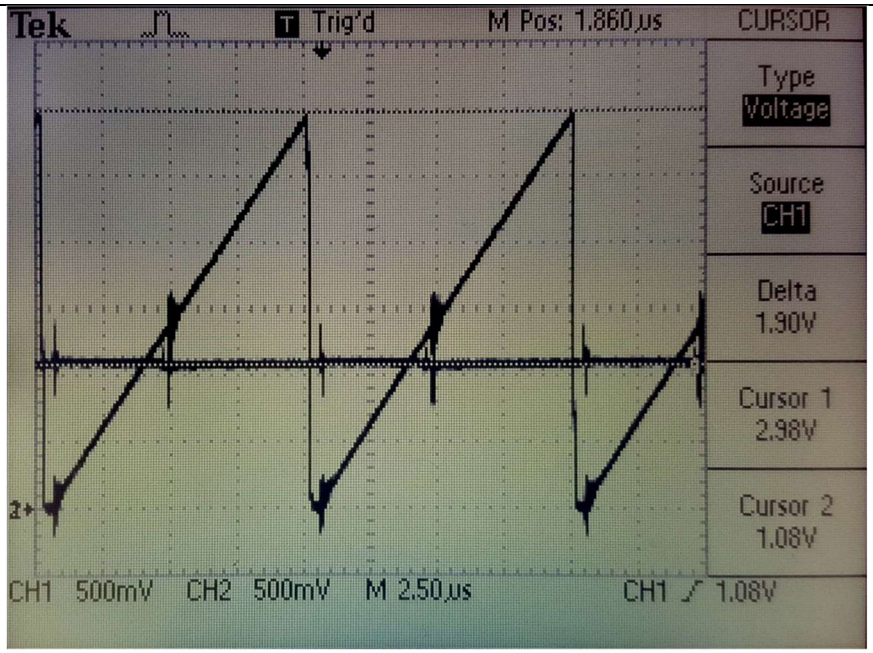

Fig. 11Saw tooth signal and controller output voltage at 14.5V input

Fig. 11 shows the saw tooth signal and the controller output voltage at an input voltage of 14.5V. The saw tooth's voltage is 2.98V as shown in "Cursor 1" block and the controller output voltage is 1.08V as shown in "Cursor 2" block, at an input voltage of 14.5V.

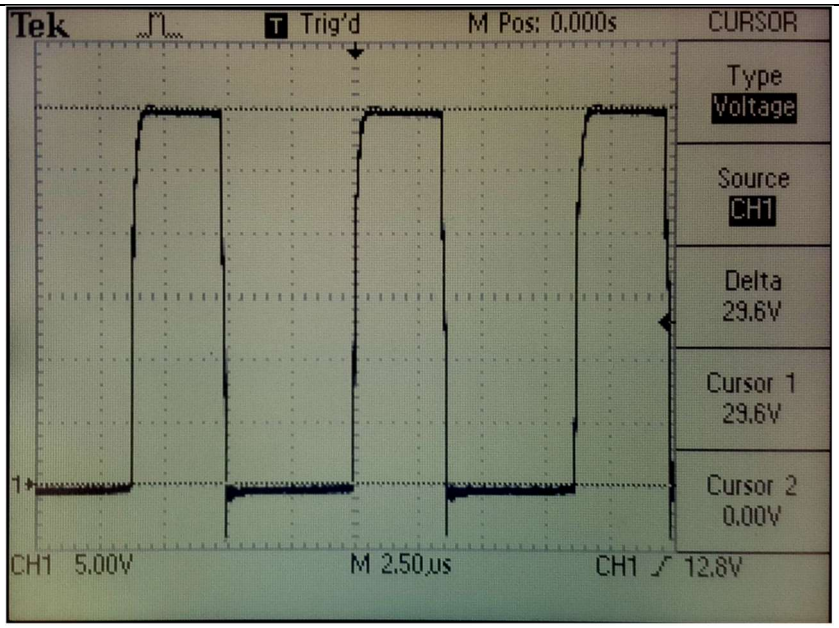

Fig. 12 Switching square wave amplitude at 14.5V input

Fig. 12 shows the switching square wave at an input voltage of 14.5V. The figure shows the amplitude of the switching square wave shown in “Cursor 1” and “Cursor 2” blocks, at an input voltage of 14.5V.

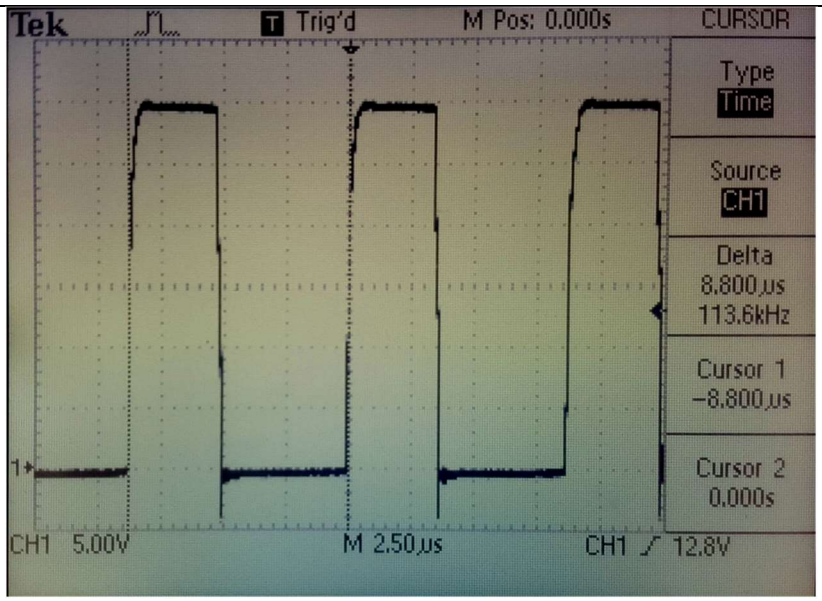

Fig. 13 Period of switching square wave at 14.5V input

Fig. 13 shows the switching square wave at an input voltage of 14.5V. The figure shows the period of the switching square wave shown in “Cursor 1” and “Cursor 2” blocks, at an input voltage of 14.5V.

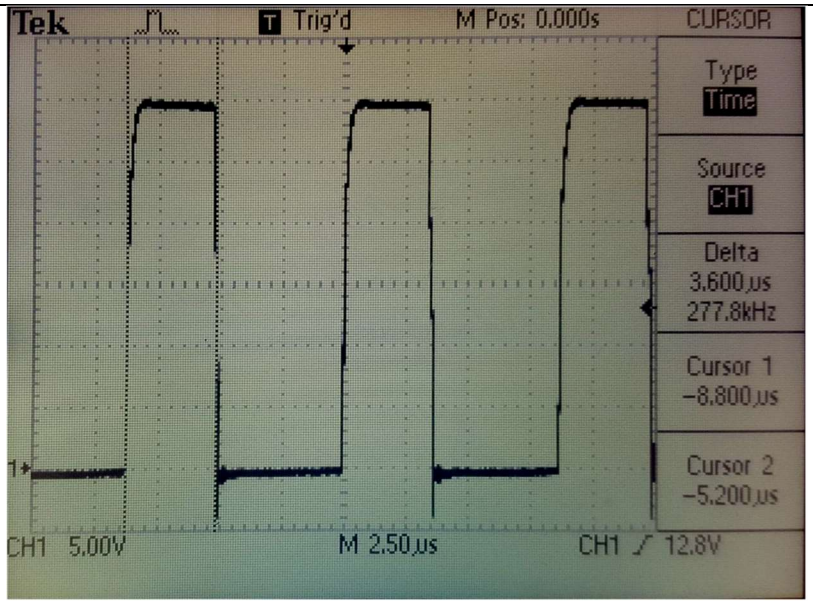

Fig. 14 On time of switching square wave at 14.5V input

Fig. 14 shows the switching square wave at an input voltage of 14.5V. The figure shows the on time of the switching square wave shown in “Cursor 1” and “Cursor 2” blocks, at an input voltage of 14.5V.

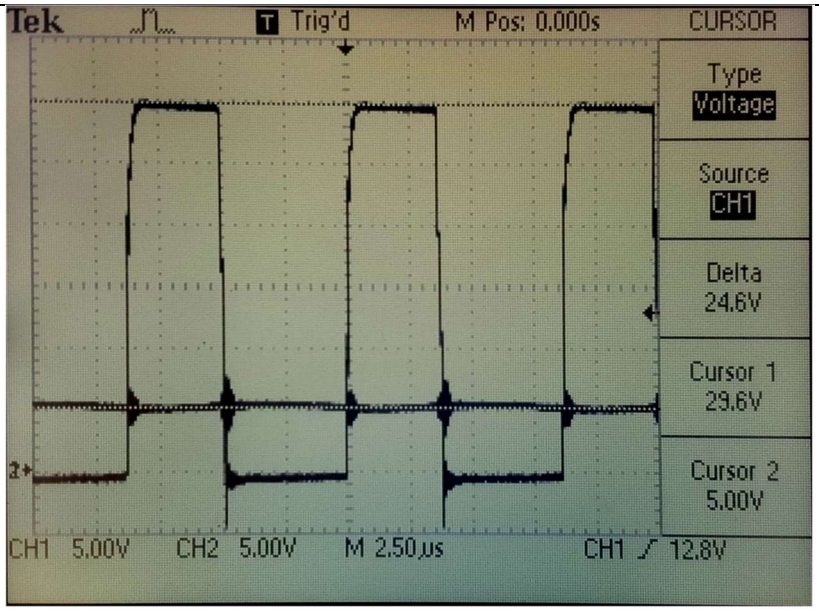

Fig. 15 Control voltage and switching square wave at 14.5V input

Fig. 15 shows the control voltage signal and switching frequency square wave at an input voltage of 14.5V. To show the duty cycle with the amplitude of the control voltage which is 5V as shown in “Cursor 2” block, at an input voltage of 14.5V.